

LAKSHYA KUMAR

Engineering Leader | AI, Web3, Distributed Systems | 8 Years

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PROFESSIONAL SUMMARY

Engineering leader, eight years across fintech, AI and Web3, most of it on early-stage builds rather than long runs on mature systems. I've shipped a Polygon and Ethereum payments platform doing 25,000+ transactions a day, an institutional fund tokenization platform on EVM and Provenance, and an enterprise RAG assistant that's now in production. Right now I'm tech lead at Payram and ZepaPay, the on-ramp/off-ramp crypto payments platform, where I own the distributed ledger and the AI agents behind support and operations. I move between architecture review and Go concurrency code without much friction, and I've taken teams from five to fifteen while staying hands-on. Core maintainer of Hyperledger Bevel, author of Mjolnir.

CORE SKILLS

Languages: TypeScript, JavaScript, Python, Go, Solidity, Rust

Architecture & Code Quality: SOLID principles, clean and layered architecture, one-way dependency rules, interface-driven design, domain modelling, design patterns, refactoring, code review standards

Backend & Systems: Node.js, FastAPI, microservices, REST, event-driven queues, concurrent worker pools, Postgres, Redis, database sharding with scatter-gather aggregation, ACID and BASE design, connection pooling, horizontal scaling

AI & Agents: LangChain, LangGraph, CrewAI, PydanticAI, RAG pipelines, hybrid retrieval (BM25 + dense, RRF), reranking, vector databases, MCP servers, tool orchestration, agent guardrails, prompt-injection defense, model routing

Evals & LLMOps: LLM-as-judge, trajectory evals, golden datasets, CI eval gates, drift detection, fine-tuning (LoRA / QLoRA, PEFT), model serving (vLLM, Triton), quantization, MLflow, Langfuse

Production-Readiness: OpenTelemetry, structured logging, RED metrics, p95/p99 SLOs, circuit breakers, distributed rate limiting, leader election, graceful shutdown, distroless containers, fuzz and property-based testing, schema migrations

Blockchain: EVM, Provenance, Hyperledger Fabric and Besu, Polygon, Avalanche, ERC-1400, ERC-4337, account abstraction, cross-chain bridges, tokenization

Cloud & DevOps: AWS (ECS, EC2, Lambda, S3, ECR), Docker, Kubernetes, autoscaling, GitHub Actions, Jenkins, GitOps, blue-green and canary deployments

Leadership: architecture review, team lead (up to 15 engineers), hiring, mentorship, delivery management, technical strategy

PROFESSIONAL EXPERIENCE

Tech Lead (Contract) | Payram & ZepaPay

Nov 2025 - Present · Independent engagement, sole architect and owner

- Tech lead for a self-hosted crypto payments platform in Go, and for ZepaPay, the on-ramp/off-ramp product built on Payram's rails for Consult Now. ZepaPay moves money between fiat and crypto, handles KYC and compliance, and settles end to end.
- Built ZepaPay's ledger as a sharded distributed store. Balance and settlement reads fan out across shards and get aggregated on the way back. No single table holds the whole account set, and adding capacity means adding a shard, not rewriting the schema.
- Built the AI agents in the payments and support flows. They call tools against strict schemas, sit behind input, output and scope guardrails, and route between models to keep cost and latency down. Nothing ships until it clears an eval suite of trajectory checks plus an LLM judge.
- Event-driven push and pull queues with concurrent workers and cron-scheduled async settlement, decoupling payment initiation from dispatch. Worker pools scale horizontally and blue-green pipelines give zero-downtime releases with sub-second dispatch under concurrent load.
- Wired OpenTelemetry across HTTP, Postgres and Redis with W3C trace-context propagation; structured JSON logs on stdout carry request and trace IDs. Cut mean detection time on the last two production incidents to under 4 minutes.
- Postgres advisory locks gate the cron settlement worker so exactly one replica runs per tick. That closed out a duplicate-charge race we hit in the first week of the rollout.
- Circuit breakers on every outbound call split infrastructure errors from client-fault errors, so a partial Redis blip in March never reached the API tier. A Redis-backed token-bucket limiter holds a 200 rps per-instance cap and fails closed on Redis errors, verified at 1,200 rps aggregate across 6 instances under load.

VP, Web3 and AI Solutions | Formidium

Feb 2024 - Oct 2025

- Grew the Web3 and AI organization from six to fifteen engineers across two product lines, owning hiring, architecture review and quarterly delivery.
- Delivered an institutional fund tokenization platform on EVM and Provenance using ERC-1400, with ACID semantics on the off-chain ledger and BASE for distributed on-chain state.
- Built an enterprise RAG assistant in Python (LangChain, LangGraph, Next.js front-end) that cut analyst research time by 60%. Retrieval combined keyword and vector search with a reranking pass, and every answer carried citations back to source for analysts to check.
- Put the assistant behind an eval suite that caught quality regressions in CI before release, scoring answers against a golden set with an LLM judge. Kept p95 latency in check with caching and streamed responses so it felt fast.
- Brought NAV update lag from minutes down to seconds through blockchain oracles and an event-driven update pipeline.
- Removed single points of failure in custody by re-architecting the wallet layer around multi-sharded key storage and AWS Secrets Manager.
- Standardised the Web3 services on a layered architecture with one-way dependencies and interfaces at the boundaries, so a storage or chain client could be swapped without touching its callers. Added OpenTelemetry tracing across HTTP, Postgres and the wallet RPC. Median incident triage moved from about 45 minutes to roughly 12 over two quarters.

- Moved Postgres schema changes to versioned, ordered migration files run as a deploy pre-hook. The "new binary on old schema" incident class went from roughly one a month to zero.

Backend Engineer | [Xalts](#)

Apr 2023 - Dec 2023

- Owned all backend and blockchain modules over a six-month delivery window while mentoring a team of five.
- Onboarded banking clients to a private EVM chain on Avalanche and Hyperledger Besu via a custodial wallet with Web2 onboarding and automated key generation.
- Used ERC-4337 account abstraction to enable programmable enterprise transaction flows through smart contract wallets.
- Implemented cross-chain transfers between Besu and Avalanche with a blockchain indexer and token bridge, cutting settlement from hours to under a minute.
- Split the test suite into a fast tier of about 340 unit tests on in-memory fakes, finishing in under 10 seconds, and an integration tier on testcontainers for Postgres and the Besu node. Both make the same target the team can run locally.

Blockchain Lead | [Quadrant.io](#)

Dec 2021 - Mar 2023

- Led a five-engineer team across multiple blockchain applications and production releases.
- Designed and built a Polygon and Ethereum platform running 25,000+ daily transactions with zero downtime, using concurrent processing, message queues, Go channels, and retry-and-recovery logic.
- Designed the platform's private EVM chain for multi-region data consistency: distributed database architecture, full-node infrastructure, bridge, explorer and APIs.
- Benchmarked the hot-path RLP encoder and caught a refactor in review that would have shipped about 3.4x slower.

Concept Engineer | [Everledger](#)

Feb 2021 - Oct 2021

- Shipped React provenance widgets for diamond authenticity verification into client-facing production applications.
- Expanded the platform into the Chinese market through a WeChat API integration and Hyperledger Fabric extensions for local compliance.

Software Engineer | [Accenture](#)

Sep 2018 - Jan 2021

- Core contributor to Hyperledger Bevel, the open-source Kubernetes framework for production blockchain deployment now used widely across enterprise.
- Built deployment pipelines for Hyperledger Fabric, Hyperledger Besu and JP Morgan Quorum with CI/CD and GitOps via Jenkins and Travis CI.

SELECTED PROJECTS

ZepaPay | [app.zepapay.com](#)

On-ramp/off-ramp crypto payments platform built on Payram's rails for Consult Now. Fiat-to-crypto conversion, KYC and compliance, a sharded distributed ledger, and settlement. I was tech lead, and built the AI agents that run behind support and operations.

Mjolinir | [github.com/lakshyakumar/mjolinir](#)

Open-source production-readiness audit, language and agent agnostic. Walks any backend service against a 19-item checklist covering circuit breakers, rate limiting, leader election, tracing, RED metrics, graceful shutdown and schema migrations, then returns a single WORTHY or NOT WORTHY verdict with file-level evidence. Ships as a skill for Claude Code and Cowork.

Capstok | [capstok.com](#)

Hands-on learning platform with 7,500+ challenges across 14 tracks spanning AI/ML, blockchain, cloud, security and systems. Claude, Cursor or any MCP-aware agent can pull the material over a read-only API to teach and quiz you, in 40+ languages.

Hyperledger Bevel | [github.com/hyperledger/bevel](#)

Open-source accelerator for distributed ledger deployment on Kubernetes. Core contributor.

Advanced RAG Assistant | [github.com/lakshyakumar/Advanced-RAG](#)

Production AI assistant for document Q&A. Rephrases and simplifies queries before retrieval; 60% faster research through LangChain and LangGraph orchestration.

RepoMind | [github.com/lakshyakumar/repomind](#)

Repository-intelligence MCP server for coding agents. Exposes structured codebase context, semantic search and file-level queries as AI-callable tools.

WRITING & OPEN SOURCE

Technical blog: [medium.com/@lklsquare](#). Account abstraction (ERC-4337), AI agent tooling, RAG systems, vector search and MCP servers.

- [Mastering GitFlow: A Production-Ready Branching Strategy for Enterprise Development Teams](#)
- [Teaching AI to Remember: Episodic, Semantic, and Procedural Memory in AI](#)
- [Vector Search in MongoDB: Best Practices for Real-Time Semantic Retrieval](#)

Open source: core contributor to Hyperledger Bevel; author of Mjolinir; maintainer of Turbulent (Node performance toolkit), Wallact (ethers.js contract-interaction wrapper) and XIToDb (Excel-to-database loader) on npm.

EDUCATION

M.Tech, Computational Mathematics | NIT Surathkal, Karnataka Jul 2016 - Jun 2018

B.Tech | Delhi Technological University Aug 2011 - Jun 2015